

Assessed Coursework Coversheet

For use with *individual* assessed work

Student ID Number:	2	0	1	6	1	7	3	0	2
Module Code:	LUBS1890								
Module Title:	Starting Your Own Business								
Module Leader:	Deema Refai								
Declared Word Count:	2500								

Please Note:

Your declared word count must be accurate, and should not mislead. Making a fraudulent statement concerning the work submitted for assessment could be considered academic malpractice and investigated as such. If the amount of work submitted is higher than that specified by the word limit or that declared on your word count, this may be reflected in the mark awarded and noted through individual feedback given to you.

It is not acceptable to present matters of substance, which should be included in the main body of the text, in the appendices ("appendix abuse"). It is not acceptable to attempt to hide words in graphs and diagrams; only text which is strictly necessary should be included in graphs and diagrams.

By submitting an assignment you confirm you have read and understood the University of Leeds **Declaration of Academic Integrity** (http://www.leeds.ac.uk/secretariat/documents/academic_integrity.pdf).

Project Synapse

The COVID-19 pandemic introduced immeasurable strain upon the NHS, revealing significant flaws within. One such major flaw concerns the plethora of non-clinical admin tasks doctors need to perform daily, taking up precious time and energy that should be for patient care. This has caused the healthcare industry, particularly healthcare staff in the NHS, to become fatigued. Project Synapse aims to create an advanced interfacing system leveraging AI technology, to record and act upon patient data with great time efficiency and little expenditure of energy.

This report will discuss the ideas and feasibility of Project Synapse, including the ability for the business to flourish within the electronic patient records (EPR) industry. The report will utilise scrutinised references from sources from IBISWorld, while using up-to-date secondary research from online articles and health-related websites to back up my findings. This report will demonstrate the competitive nature of the project via the “Blue Ocean Strategy” concept.

Project Synapse is a concept derived from desperation, deprived morale, and essential need. Building this system can help to revitalise healthcare as an industry, preferring patient care over bureaucracy. By these factors, Project Synapse will naturally come with widespread adoption across the nation, leading to a successful business.

Contents

Project Synapse	2
Entrepreneur Feasibility Analysis.....	4
Market Feasibility Analysis	5
Industry Feasibility Analysis	6
Blue Ocean Strategy	8
Conclusion	9
Appendix.....	10

Entrepreneur Feasibility Analysis

My current education involves studying computer science with artificial intelligence in an integrated masters program. The degree offers me a lot of skills in relation for this project, specifically skills in project management of relatively large scale, developing AI systems using cutting-edge algorithms and data science techniques, and excellent engineering/programming principles and agile methods of development. I have excellent advanced background knowledge in many STEM related subjects, including but not limited to maths, physics, chemistry and biology. I have cultivated my knowledge within the technology, biotechnology, healthcare and AI sectors (including studying patient records and diseases) during my time in lockdown during Covid-19. Knowing all this, plus my undying resilience and attention to detail, gives me the best idea on how to create Project Synapse, making me the best person to lead the development.

My interest in healthcare technology industry stems from my decision to study computer science at degree level, where I found this choice could allow me to go into any field of my choosing. My aspirations fall towards the use of AI and robotics within the medical industry, whether it involves discovering the next best cure to cancer, to developing and deploying assistive, complex technologies similar to Big Hero 6's Baymax. With today's emergence of generative AI such as ChatGPT, such dreams all of a sudden became a truly possible reality.

When the nation emerged from the aftermath of the Coronavirus pandemic, it was clear the healthcare system was beaten and broken. Covid-19 revealed to me how fragile the current NHS is, underfunded and short on hospital beds for those critically ill. People died waiting in queues for clinicians under heavy workload pressure and low incentives, where morale took a significant hit. Then there were the doctors strikes, where pay (and therefore the NHS budget) became stretched to the point of being unsustainable for the workforce. This was when I monitored the current status of the NHS, critically analysing any areas for improvement across the board. Coincidentally, generative AI such as ChatGPT was launched, leading me to study this technology, its processes and its limitless applications in the healthcare sector.

It was at this point I conceived Project Synapse, thoroughly thinking through the features and applications for the requirements of today. My critical analysis and creative thinking, adaptive processing and logical reasoning makes me the most adept and capable to this task. Because of my value in seeing Project Synapse as a mission for a national cause, my loyalty to the project is unwavering and my insight to the design elements is invaluable. I will do everything in my power to ensure that project is completed to the best of my ability.

Market Feasibility Analysis

The NHS is in a state of crisis. With an already hit morale coming out of the Covid-19 pandemic and underfunded staff and facilities, hospital workers across the nation are under immense strain due to the amount of paperwork staff have to fill in (Caldwell, 2023). This paperwork can range from patient-clinician interactions to follow up tests and referrals, to surgery reports and ward rounds (Harvard Health, 2016). Essentially, healthcare services document everything about the patient from step 1 to the end. This clerical work is primarily being done by the junior doctor workforce, who are already overburdened with strenuous workloads and little pay (Preston, 2023). This became a healthcare-wide problem following the junior doctors strikes, where because there were no junior doctors available to document these records, the senior staff had to step in. This costed the NHS valuable expert-level care and time (Kashif-Khan & Dawrant, 2023).

The Government and the NHS have both announced plans to radicalise healthcare by the digitalisation of key infrastructure points, via the adoption of EPR's and other digital systems. These plans come with an investment fund of £2 billion to achieve this goal (HM Government, 2022), making the industry an enticing endeavour in terms of financial support and greater profitability proportions.

Current forms of documentation software involve using outdated methods of note-taking via pen and paper (Dhaliwal, n.d.), typing notes up on the computer, or using speech to text software to record the interaction between patient and doctor. All methods mentioned are ineffective solutions, costing roughly the same amount of time or more to complete (Harvard Health, 2016).

Because of the scale of the NHS, the scope can be defined within (but not limited to) the UK and its healthcare sectors, whether it be public or private (since doctors are working in both areas). This amounts to approximately 1.8 million people (Statista, 2022), and over 200 health service providers across the nation, which can pose as a significantly large target market (Anon, n.d.). Approximately 1.6 billion transactions of patient data are conducted per month on the NHS Spine network (NHS Digital, n.d.), making it possible to create a business model profitable on a small fee per transaction. We can expect this target market to grow as these healthcare services tend towards adding on to their existing digital technologies, to meet increasing demands for digitisation of patient medical records (Ltd, n.d.).

There are three approaches to marketing Project Synapse, either directly to the medical community as a personal app, or by targeting healthcare providers to take up the system within their scope of operations, or a combination of both. Because of the number of people under burden, the opportunity presents itself as one of need and desperation. As such, the product has the potential to be marketed and taken up by the medical community very quickly, despite the different salaries for junior and senior doctors. There is also the opportunity to market this product to become more cost efficient for the targeted industry compared to existing systems by eliminating the time wasted on these admin tasks.

As for the competition within the market sector, there is very limited evidence to suggest anyone is actively working to seize this opportunity. As it currently stands, many companies appear to be using dated technology, which receives limited updates to the user interface and additional functionality. As a result, innovation in this specific market is low.

Industry Feasibility Analysis

The target industry to operate in, and the closest related target industry in relation to the project, will concern the electronic patient records (EPR) industry in the UK. Because of the limited reporting on this highly specialised industry, this section will be covered using free versions of reports (Htn, 2020), (Priestman et al., 2018) and (Knowledge Sourcing Intelligence LLP, 2023), and directly from (HM Government, 2022).

The electronic patient record (EPR) industry involves creating software to store, receive and manage patient data safely and securely. These systems play an integral part of ensuring medical diagnosis is conducted accurately, including but not limited to past and recurring medical conditions such as seizures, family history records, recent hospital trips and surgical/follow up work. EPR's are used by doctors, clinicians and nurses to administer health services and can also be used for research purposes via pseudonymisation of data (something which the projects' AI systems can take advantage of).

As from July 2020, EPR systems branch to 2 major types:

- Best of breed (37% of market share) – focusing on the adoption of a best system for a specific task
- Integrated EPR's (56% of market share) – focusing on 1 system based on a single central database, rolled out to the whole organisation.

Integrated EPR's are becoming the industry norm, with these systems slowly being integrated into the NHS over a span of 20 years. It is believed integrated EPR's can help reduce the overall costs for the NHS and unwanted variation to improve quality in care and safety. Given these current trends, it is safe to assume Project Synapse will take the approach of building an integrated EPR, diving into an emerging yet competitive market segment.

In terms of the scale of the industry, the UK's EPR market has an estimated value of \$1 million in 2021 and is expected to grow by 11% to \$2 million by 2028. Demand for new systems to record patient data is growing due to the NHS's commitment to making patient records paperless via digitisation, and the governments increase in healthcare expenditure. Industry giants involve companies such as Cerner, System C, Lorenzo and Epic, who take up around 70% of current contracts issued to healthcare providers. These companies are developing their EPR systems with capabilities in electronic prescription, medical results, patient management, data collection and many more, making their systems effective at completing certain tasks (Patel and Jones, 2021), and highly competitive propositions for trusts looking to invest in such an EPR system.

Current trends within the EPR industry involve some of the following (Healthcare IT News, 2020):

- Increase in cloud-based segment growth. Cloud computing allows for easy remote access to patient data, reducing overall costs and increasing security.
- Integration with business intelligence (BI) to help make data-driven decisions, enhance operational efficiency and support financial stability
- Integration with the telehealth industry, an emerging industry in the UK concerned with patient consultations over the phone. EPR systems help to retrieve patient data quickly and efficiently, while reducing administrative tasks.

There are some open opportunities to exploit here, since companies creating these EPR systems do not use the latest advances in generative AI. By employing AI into EPR systems, AI can help to create and automate tasks and documentation, while creating a streamlined and nuanced system. Also, implementation of these systems has been slow due to the necessity of standardisation against the governments healthcare databasing system and compliance in data transfer. There have been additional operational and behavioural barriers for successful integration, including the poor usage of these systems to their full extent by doctors due to workflow changes from paper to electronic, another area to take advantage of.

Purchase and integration of corporate EPR systems works by contract. Here, there may be an opportunity to slip into the industry. This gap is swiftly closing however, with the current industry transformation reaching the endgame of 90% in 2023 of all NHS providers achieving an operating EPR system. This may indicate the need to shift the market towards those working in the NHS rather than targeting these healthcare trusts, until a mature, feasible and efficient product becomes available for corporate use, and when these contracts begin to end.

There are significant threats within this industry, 1 major one concerning the competitor Cerner, a subsidiary owned by Oracle. Oracle, being a databasing company, would have a proficient data science team who may be able to quickly develop their own datasets to train AI models for their EPR systems, posing as a major competitor. There is also the threat of trying to secure clinician and doctor involvement as beta testers for the software, and the issues with complying with governmental and national policies (Clarke et al., 2015).

Blue Ocean Strategy

Project Synapse is a revolutionary EPR designed for use by, and marketed to, doctors across the NHS as a personal subscription service. This EPR will use the latest advancements in AI technology to streamline typical admin-related tasks, while providing updated interfacing technology using cloud distributed services such as speech recognition to help understand patient-doctor interactions and document them. The EPR's AI will commit to employing action items, such as setting patient tests, writing referrals, and creating/modifying patient information to store on the system. Ideas include:

Create:

- Documentation and reports automation
- Action items automation
- AI generated patient briefing packages

Raise:

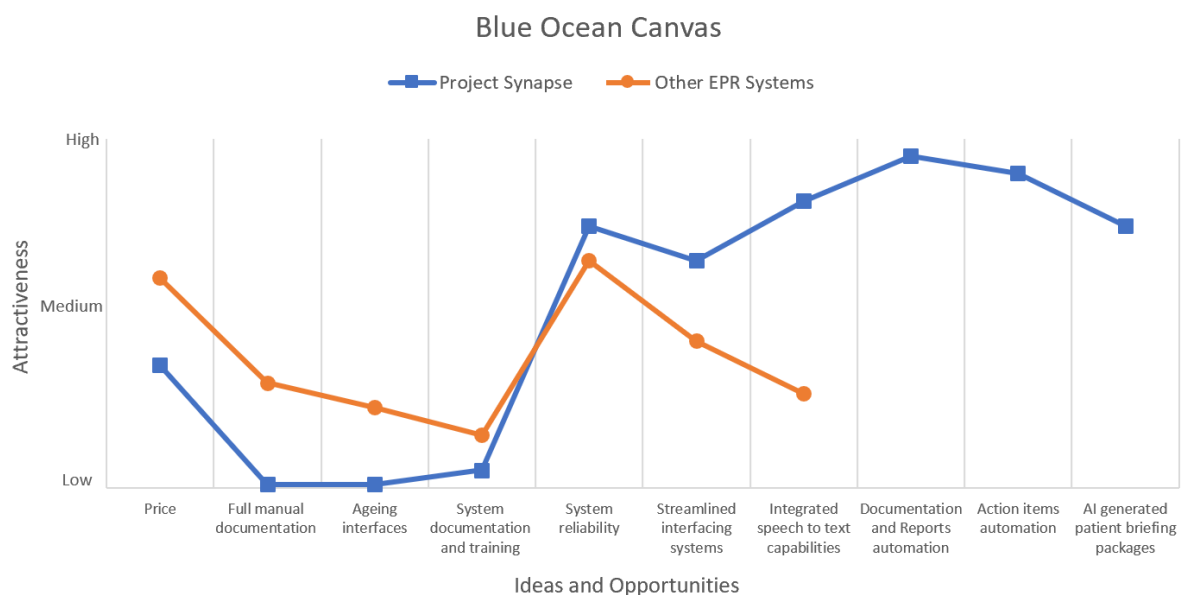
- Integrated speech to text capabilities
- Streamlined interfacing systems
- System reliability

Reduce:

- Competitive pricing
- System documentation and training, since the AI can be programmed to function as the trainer

Eliminate:

- Full manual documentation (review and editing will still be available)
- Ageing interfaces



Conclusion

This report has adequately concluded that the ideas, opportunities and proposals from each analysis section, are indeed feasible, but have been limited with some constraints. The report has identified the entrepreneur, the individual constructing this report, indeed has the skills and qualifications to create such an intricate software, with their combined knowledge and specialisations in medicine and artificial intelligence. The entrepreneur has relative experience from observing documentaries and experiencing the catastrophic NHS system to understand the extent of the issues, where they exist, and what can be done within the short timescale to try and relieve the NHS of such pressures.

The report also concludes the market and industry feasibility analysis is indeed accurate to a national extent, stressing the needs and desires of those currently operating at the front lines of the NHS. Because of a number of extraordinary events, namely the pandemic, the doctors strikes, and an ever increasing ageing population, it is clear demand for Project Synapse is at an all time high, creating an opportunity ripe for exploitation. The report has accounted for the current trends and shifts in the industry to conclude a “software as a service” model marketed directly for doctors is indeed the best approach to tackling said opportunity.

This report finally concludes by discussing the highly competitive nature of Project Synapse. Project Synapse aims to target a desperate market, with cutting edge features leveraging the latest advancements in AI. The proposed solution, with its advancements in automation and streamlined service, aims to directly tackle this problem head on, introducing these “create” and “raise” factors within the blue ocean canvas, as key success factors for the business and its customers.

Appendix

Caldwell, G. 2023. The NHS is drowning in paperwork. *The Spectator*. [Online]. Available from: <https://www.spectator.co.uk/article/the-nhs-is-drowning-in-paperwork/#:~:text=or%20legal%20case,-.In%20many%20hospitals%20as%20many%20as%2030%20pages%20of%20nursing,form%20requiring%20patient%20identification%20labels.>

Clarke, A., Adamson, J., Sheard, L., Cairns, P., Watt, I. and Wright, J. 2015. Implementing electronic patient record systems (EPRs) into England's acute, mental health and community care trusts: a mixed methods study. *BMC Medical Informatics and Decision Making*. **15**(1).

Dhaliwal, J. n.d. EPR Transformation. *KPMG*. [Online]. Available from: <https://kpmg.com/uk/en/home/industries/infrastructure-government-healthcare/healthcare/digital-health/electronic-patient-record-transformation.html>.

Harvard Health 2016. Physicians, paperwork, and paying attention to patients. *Harvard Health*. [Online]. Available from: <https://www.health.harvard.edu/blog/physicians-paperwork-and-paying-attention-to-patients-2016103110558>.

Healthcare IT News 2020. Leeds and York Partnership NHS FT goes live with new EPR system. *Healthcare IT News*. [Online]. Available from: <https://www.healthcareitnews.com/news/emea/leeds-and-york-partnership-nhs-ft-goes-live-new-epr-system#:~:text=Leeds%20and%20York%20partnership%20NHS%20FT%20has%20gone%20live%20with,by%20Advanced%20in%20November%202019>.

HM Government 2022. A plan for digital health and social care. *GOV.UK*. [Online]. [Accessed 11 December 2023]. Available from: <https://www.gov.uk/government/publications/a-plan-for-digital-health-and-social-care/a-plan-for-digital-health-and-social-care>.

Htn 2020. HTN Trends Series: EPR Trends in the UK - htn. *Htn*. [Online]. Available from: <https://htn.co.uk/2020/11/13/htn-trends-series-epr-trends-in-the-uk/>.

Knowledge Sourcing Intelligence LLP 2023. UK Electronic Medical Record (EMR) market Size: Report, 2023-2028. *Knowledge Sourcing Intelligence LLP*. [Online]. [Accessed 10 December 2023]. Available from: <https://www.knowledge-sourcing.com/report/uk-emr-electronic-medical-record-market>.

Ltd, R. and M. n.d. United Kingdom Electronic Medical Record (EMR) Market - Forecasts from 2023 to 2028. *Research and Markets Ltd 2023*. [Online]. Available from: [https://www.researchandmarkets.com/reports/5602455/united-kingdom-electronic-medical-record-emr?utm_source=BW&utm_medium=PressRelease&utm_code=8pms62&utm_campaign=1721174+-+United+Kingdom+Electronic+Medical+Record+\(EMR\)+Market%3a+Expected+to+Surge+Due+to+the+Increasing+Digitization+in+the+Country%27s+Healthcare+Sector&utm_exec=como322prd](https://www.researchandmarkets.com/reports/5602455/united-kingdom-electronic-medical-record-emr?utm_source=BW&utm_medium=PressRelease&utm_code=8pms62&utm_campaign=1721174+-+United+Kingdom+Electronic+Medical+Record+(EMR)+Market%3a+Expected+to+Surge+Due+to+the+Increasing+Digitization+in+the+Country%27s+Healthcare+Sector&utm_exec=como322prd).

NHS Digital n.d. Home - NHS Digital. *NHS Digital*. [Online]. [Accessed 10 December 2023]. Available from: <https://digital.nhs.uk/>.

Patel, A. and Jones, A. 2021. Your guide to Electronic Patient Records (EPR) | SoR. *SoR*. [Online]. Available from: <https://www.sor.org/news/information-management-technology/your-guide-to-electronic-patient-records-%28epr%29>.

Preston, H. 2023. Two months in the life of a junior doctor. Available from: <https://business.leeds.ac.uk/research-wbrc/dir-record/research-blog/2089/two-months-in-the-life-of-a-junior-doctor>.

Priestman, W., Sridharan, S., Vigne, H., Collins, R., Seamer, L. and Sebire, N.J. 2018. What to expect from electronic patient record system implementation: lessons learned from published evidence. *BMJ Health & Care Informatics*. **25**(2), pp.92–104.

Statista 2022. Number of people employed in the health and social sector UK 2022, by enterprise size. *Statista*. [Online]. Available from: <https://www.statista.com/statistics/678587/united-kingdom-uk-health-and-social-work-employment-by-number-of-employees/>.

Kashif-Khan, A.E. and Dawrant, M. (2023) 'Consultant's Opinions on Doctors Strikes'.